Changes compared to the second edition

In order to help readers and instructors familiar with the second edition, I list in the following the major changes

Chapter 2nd ed.	Chapter 3rd ed	Major changes
1	1	Updates and (minor) reorganizations on history, applications, requirements
2	2	Updates on spectrum usage and economic and behavioral aspects
3	3	Link budgets are partly updated to give examples from modern systems. The rest of Chapter 3 is new (overview of wireless systems). Noise figure discussion moved to new Chapter 17
4	4	Added discussion of outdoor-to-indoor penetration, atmospheric attenuation; former Sec. 7.5 (deterministic modeling) is updated and integrated as Sec. 4.7
5	5	Small updates
6	6	Extended discussion of directional channel characteristics (Sec. 6.6)
7	7	Completely rewritten, and considerably extended in scope.
8	9	Extended description of high-resolution parameter estimation (Sec. 9.6). Antenna arrays moved to new chapter 8. Added some new measurement methods in Sec. 9.3.
9	8	Completely rewritten and considerably extended. Added (among others) (i) fundamental description of how antennas radiate, (ii) equivalent circuit and matching, (iii) horn- and lens antennas, (iv) mm-wave antennas. Antenna arrays moved to here.
10		Eliminated (a new overview of transceiver structure is now found in Chapter 3)
11	10	Significantly reorganized, according to the structure: fundamentals – QAM – Multi-pulse modulation.
12	11	Smaller updates
13	12	Smaller updates. TX diversity moved to new Chapter 16
14	13	Significantly expanded. Added sections on: (i) small blocksize codes, (ii) BCJR algorithm, (iii) extended Turbo coding discussion, (iv) Polar codes, (v) Comparison of near-capacity-achieving codes, (vi) ARQ
15	24	Unchanged
16	14	Smaller updates
17	18	MAC contributions moved to new Chapter 18, with expanded description of OFDMA and CSMA Sec. 17.6 moved to new Chapter 21.1
18	19	Multi-user detection moved to new Chapter 28; cellular aspects of CDMA moved to Chapter 21

19	15	Added sections on (i) synchronization, (ii) adaptive modulation, (iii) generalized multicarrier techniques, (iv) OTFS
20	16	Reorganized and expanded Sec. 20.1.1-20.1.6. Considerably expanded discussion on reciprocity and feedback. Expanded discussion of 20.2, including line-of-sight MIMO, capacity of spatial multiplexing. New sections on (i) sphere decoder, (ii) hybrid beamforming, (iii) intelligent reflective surfaces, (iv) spatial modulation, (v) orbital angular momenta. Multi-user MIMO (old Sec. 20.3) moved to new Chapter 22
21	26	Smaller updates on dynamic spectrum access and UWB
22	27	Sec. 22.1-22.3, 22.6 are now Sec. 27.1-27.4. New Sec. 27.5 on network coding Sec. 22.4-22.5 are now part of Chapter 23 on ad-hoc networks
23	25	Updated with modern video codecs
24	App. 30.A	Unchanged, but moved to Appendix
25		Deleted
26	App. 30.B	Unchanged
27	31	Greatly expanded description within Sec. x.1-x.5, and added new sections on carrier aggregation, M2M communications, relaying, sidelink
28		Deleted
29	33	Expanded description of MAC protocols; added Sections on 802.11ac and 802.11ax
30	36	New examples reflecting the added material
	17	New chapter on hardware
	20	New chapter on scheduling
	21	New chapter on cellular principle (part of Sec. 21.1 is old Sec. 17.6)
	22	New chapter on multi-user MIMO, massive MIMO, Cooperative Multipoint, and cell-free massive MIMO (contains some material from old Sec. 20.3)
	23	New chapter on ad-hoc networks (routing algorithms are based on old Secs. 22.4-22.5)
	28	New chapter on interference treatment (Sec. 28.1 is old Sec. 18.4)
	29	New chapter on localization
	30	New chapter on principles on standardization
	32	New chapter on 5G NR
	34	New chapter on Bluetooth and Zigbee (802.15.4)
	35	New chapter on Beyond 5G